

Production & Costs

Basic Cost Concepts

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- 
- ▶ **Opportunity costs:** The cost of a good measured by the alternative uses that are foregone by producing the good.
- ‡
- ▶ **Accounting cost:** The actual cost paid for inputs, or alternatively the costs for which receipts are available.
- ▶ **Economic cost:** The amount required to keep an input in its present use or the amount that input would be worth in its next best alternative use.
- ▶ Includes opportunity costs. Example - inputs not paid for.
 - ▶ Does **NOT** include sunk costs.

Example

Difference between accounting and economic costs

Suppose you own a beach house that you typically rent out for \$100 a day. You are deciding whether to go on a vacation to the beach house or not for the weekend (2 days). Suppose if you were to go for the vacation, then the expenses would be: travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400). What are accounting and economic costs of the vacation?

$$\text{Acc. Cost} = 800 + 400 = 1200.$$

$$\begin{aligned}\text{Ec. Cost} &= \text{Ac. Cost} + \text{Opp. Cost} \\ &= 1200 + (2 \times 100) \\ &= 1400.\end{aligned}$$


Example

Difference between accounting and economic costs

Suppose you own a beach house that you typically rent out for \$100 a day. You are deciding whether to go on a vacation to the beach house or not for the weekend (2 days). Suppose if you were to go for the vacation, then the expenses would be: travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400). What are accounting and economic costs of the vacation?

Accountant's version of costs: They would infer that the cost of such a vacation would be a total of \$1200, which consists of travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400).

Economists' version of costs: They would estimate that the total cost as \$1400. This would include the *opportunity cost* of not being able to rent out the beach house for two days (friday-sunday), which is equal to the potential money that could have been earned if it had been rented out for that period (= \$200).

Labor Vs. Capital Costs

Costs of different inputs in production

2 inputs.

$$Y = f(X_1, X_2)$$

labor. $\swarrow$ L , $\searrow$ K $\swarrow$ Capital
machinery. $\swarrow$ (money)

Labor Vs. Capital Costs

Costs of different inputs in production

Labor Costs

To the accountant, firms spending on wages and salaries is a current expense.

Economists regard wage payments as an *explicit* cost, wherein labor services are purchased at a cost (wage rate).

Both economists & accountants view this **similarly**.

Labor Vs. Capital Costs

Costs of different inputs in production

Labor Costs

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Capital Costs

Accountants use historical price of capital and apply depreciation rules.

Economists calculate the *implicit* cost of capital by focussing on what others might pay to use it (rental rate).

Economists & accountants view this **differently**.

Handwritten notes in green ink:
 - A circled "20,000" with a dollar sign (\$) next to it.
 - A circled "2000" below it.
 - "10 years" written to the right.

Total Costs

$$TC = \text{cost of } L + \text{cost of } K$$

$$= \underbrace{w \cdot L}_{\text{wage}} + \underbrace{v \cdot K}_{\text{rental rate}}$$

→ $\underbrace{wL + vK = TC}_{\text{straight line}}$ numbers/constants.

Total Costs

Assumptions:

- ▶ There are two inputs: Labor (L) and Capital (K).
- ▶ The input prices are given and constant.

Given these assumptions, total cost (TC) can be expressed as:

$$TC = \underbrace{wL}_{\text{Labor Costs}} + \underbrace{vK}_{\text{Cost of Capital}}$$


Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

$$\underbrace{TC}_{\text{fix}} = \underbrace{wL}_{\text{numbers}} + \underbrace{vK}_{\text{numbers}}$$

Straight
line

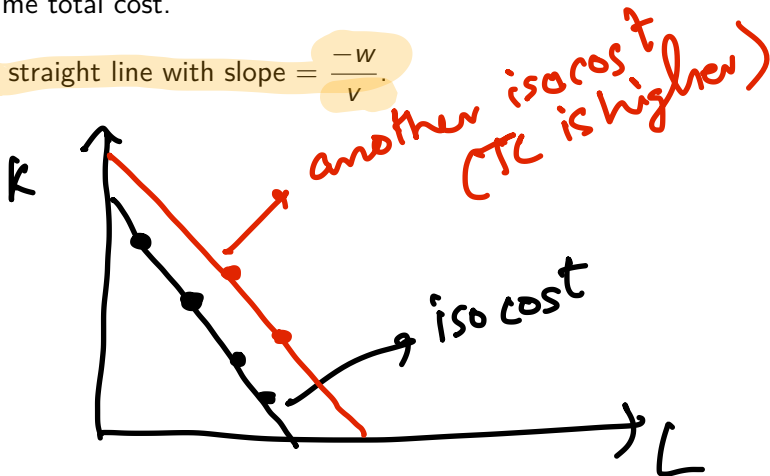
$$\underbrace{\text{constant} = wL + vK}_{\text{slope} = -\frac{w}{v}}$$

Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

It is a straight line with slope = $-\frac{w}{v}$.



Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

It is a straight line with slope $= \frac{-w}{v}$.

This is akin to the budget constraint. (Think why!)

Economic Profits $\rightarrow$ max. Profits.

$$\text{Profits}(\pi) = \text{Total Revenue} - \text{Total Costs}$$

$$= P \cdot q - TC$$

$$= P \cdot q - [wL + rK]$$

$$= Pq - wL - rK$$

Economic Profits

$$\text{Profits}(\pi) = \text{Total Revenue} - \text{Total Costs}$$

If the firm's production function is given by $q = f(K, L)$, then the total revenues of the firm can be expressed as $P.f(K, L)$, where P is the per-unit price of the good produced (output). Therefore, profits (π) can be calculated using:

$$\text{Profits}(\pi) = P.f(K, L) - wL - vK$$

Profit maximization

Given a particular output level

Known: w, v, q_1

Unknown: L, K

$$\text{Profits}(\pi) = P \cdot f(K, L) - wL - vK$$

We start with the assumption that the firm assumes a certain output level (say q_1). Then,

Profits =
(maximize)

$P \cdot q_1$
Constant

$- wL - vK$
minimized

Profit maximization

Given a particular output level

$$Profits(\pi) = P \cdot f(K, L) - wL - vK$$

We start with the assumption that the firm assumes a certain output level (say q_1). Then,

$$Profits(\pi) = P \cdot q_1 - wL - vK$$

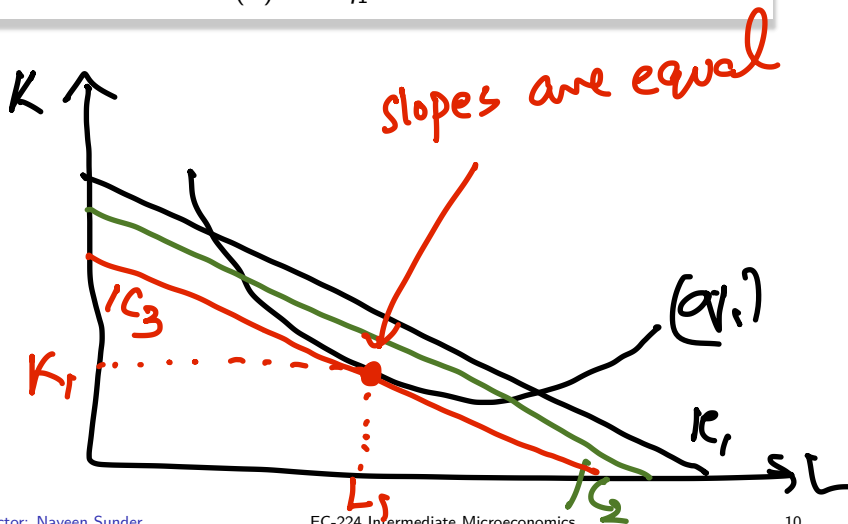
Therefore, given an output level of q_1 , the firm can maximize profits by minimizing costs. The profit maximization problem can be seen as a cost minimization one.

Cost Minimization

Graphical approach

isocost : min. cost

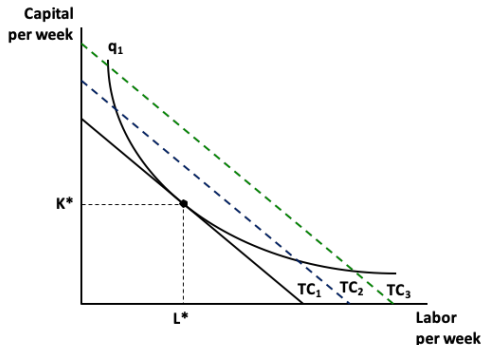
$$\text{Profits}(\pi) = P \cdot q_1 - wL - vK$$



Cost Minimization

Graphical approach

$$Profits(\pi) = P \cdot q_1 - wL - vK$$



The cost minimization problem is graphically equivalent to finding the (K, L) combination on the isoquant for quantity q_1 where the cost of production is minimized.

Cost Minimization

Optimality condition

$$\underbrace{\text{Rate of technical substitution (RTS) of L and K}}_{\text{-Slope of Isoquant}} = \underbrace{\frac{-w}{v}}_{\text{Slope of isocost}}$$

where w = wage rate (per unit cost labor cost),
and v = per unit cost of capital

Cost Minimization

Optimality condition


$$\frac{f_{MP_L}}{MP_K} = \frac{fw}{v}$$

Rate of technical substitution (RTS) of L and K = $\frac{-w}{v}$
-Slope of Isoquant Slope of isocost

where w = wage rate (per unit cost labor cost),
and v = per unit cost of capital

Equilibrium: $\frac{MP_L}{MP_K} = \frac{w}{v}$

Cost Minimization

"Bang for the buck" interpretation

Benefit of 1
move L

$$\frac{MP_L}{w} = \frac{MP_K}{v}$$

Benefit of one
move K

cost of one
move L

cost of one
move K

Cost Minimization

"Bang for the buck" interpretation

$$\frac{MP_L}{w} = \frac{MP_K}{v}$$

This condition implies that the producer should use K and L in such a way that they receive the same "bang for the buck".

Explanation: This approach can alternatively be thought of as equating the "per-dollar" benefit of additional labor with that of additional capital. The gain from using one more unit of labor is the additional output produced by it, which is equal to MP_L . The costs of using one more unit of labor are the wages paid, which is equal to w . Therefore, the "per-dollar" value of this additional unit of labor is $\frac{MP_L}{w}$. The same is the case for capital.

Total Cost function

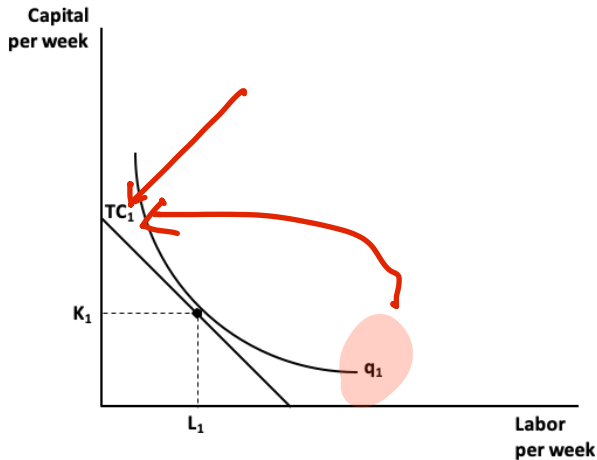
Suppose we start with a production level of q_1

$$TC = wL + vk$$

$q_1 \xrightarrow{\text{min. cost}} TC_1$

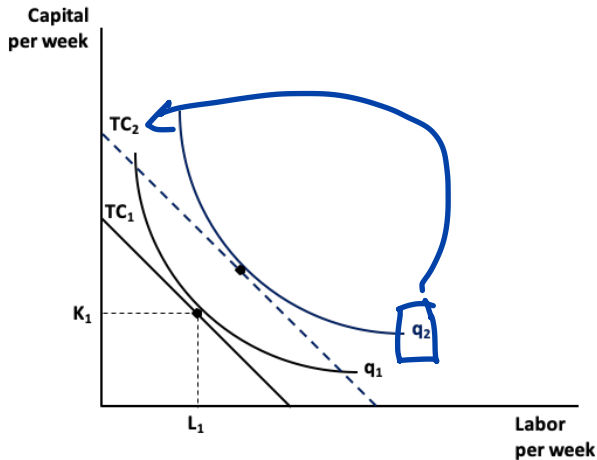
Total Cost function

Suppose we start with a production level of q_1



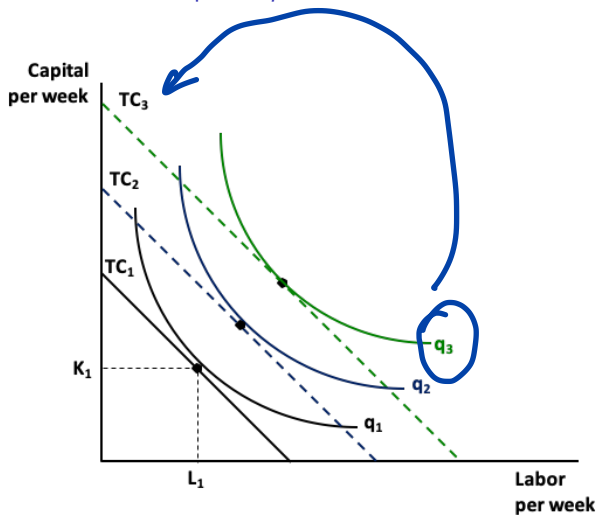
Now we want to shift to output level q_2

Then do cost minimization for output = q_2



Now we want to increase output to q_3

Then do cost minimization for output = q_3

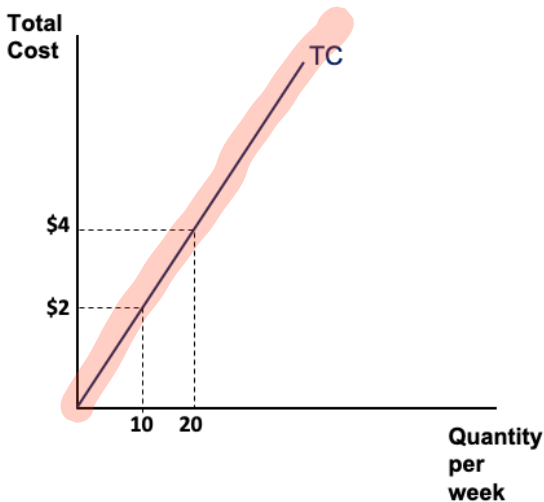


Firm's total cost (TC) curve

In the current case, the output and costs increase at the same rate
- thus this is a situation with constant returns to scale.

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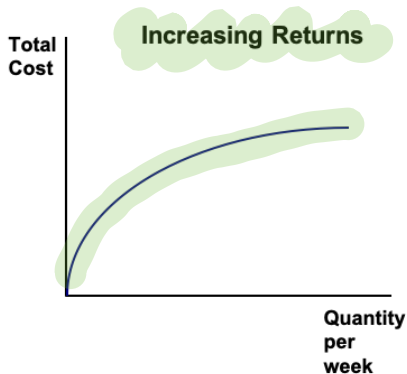


Firm's total cost (TC) curve

Increasing/Decreasing Returns

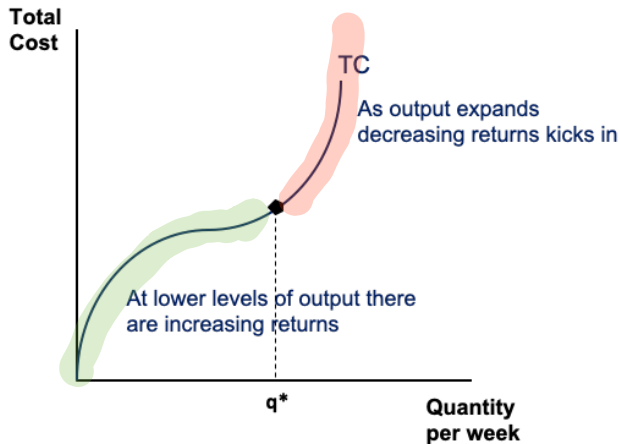
Firm's total cost (TC) curve

Increasing/Decreasing Returns



Firm's total cost (TC) curve

Firm with efficient scale – combining most scenarios.



Average and Marginal costs

Average and Marginal costs

$$\text{Average cost (AC)} = \frac{\text{Total Cost}}{\text{Quantity}} = \frac{TC}{q}$$

This is the per-unit cost of production.

Average and Marginal costs

$$\text{Average cost (AC)} = \frac{\text{Total Cost}}{\text{Quantity}} = \frac{TC}{q}$$

This is the per-unit cost of production.

$$\text{Marginal cost (MC)} = \frac{\text{Change in Total Costs}}{\text{Change in Quantity}} = \frac{\Delta TC}{\Delta q}$$

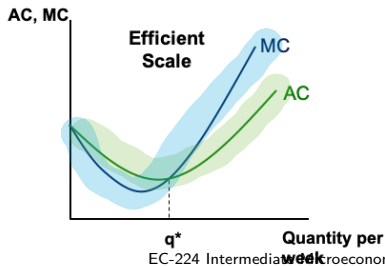
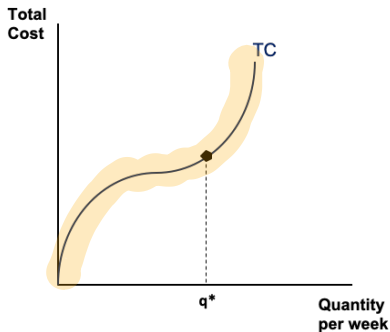
Marginal cost is the additional cost of producing one more unit of output. It is represented by the slope of the total cost curve.

Firms with efficient scale

Shape of AC and MC curves

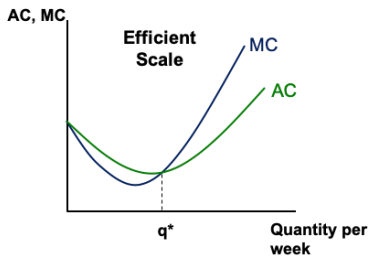
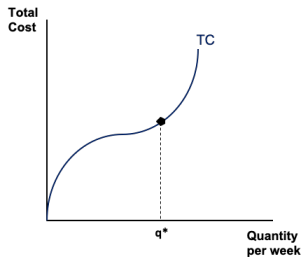
Firms with efficient scale

Shape of AC and MC curves



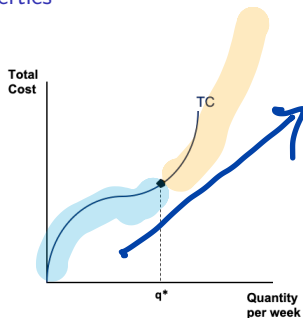
Firms with efficient scale

Properties

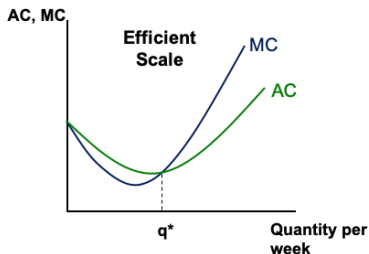


Firms with efficient scale

Properties

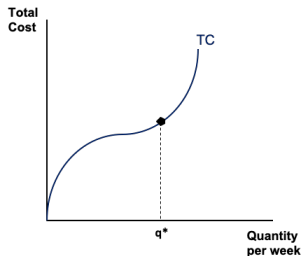


The total costs are always rising, first at a decreasing rate (concave), and then at an increasing rate (convex).

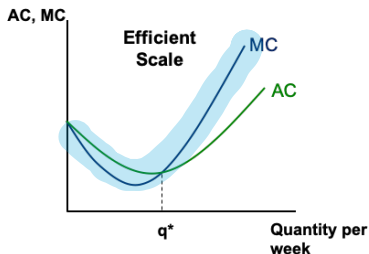


Firms with efficient scale

Properties



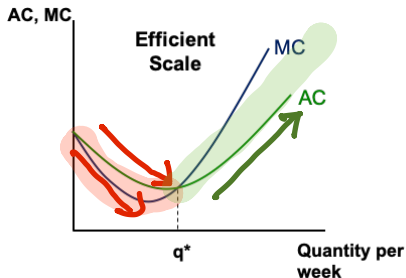
The total costs are always rising, first at a decreasing rate (concave), and then at an increasing rate (convex).



MC (slope of TC) is always positive – it falls till a certain point and then rises.

MC-AC relationship

Firms with efficient scale



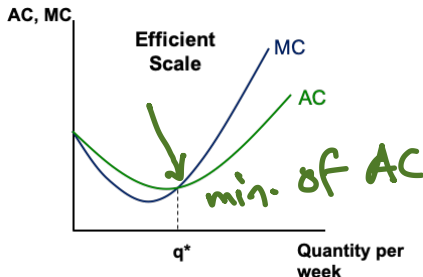
69 80
When $MC < AC$, then $AC \downarrow$

The cost of each additional unit is low, and hence the AC of increasing output is falling.

$MC > AC$, then $AC \uparrow$
93 80

MC-AC relationship

Firms with efficient scale



When $MC < AC$, then $AC \downarrow$

The cost of each additional unit is low, and hence the AC of increasing output is falling.

When $MC > AC$, then $AC \uparrow$

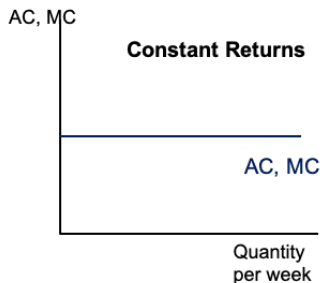
The cost of each additional unit is high, and therefore the AC of increasing output is rising.

Average and Marginal costs

Different scenarios

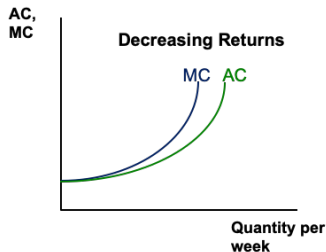
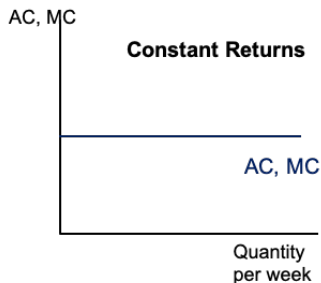
Average and Marginal costs

Different scenarios



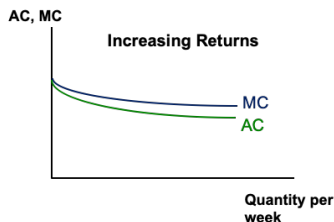
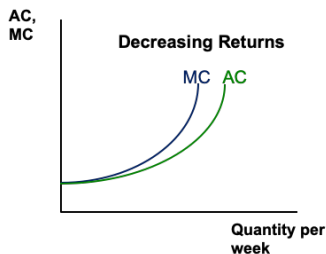
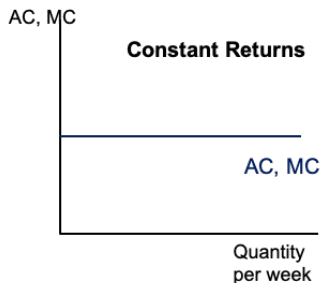
Average and Marginal costs

Different scenarios



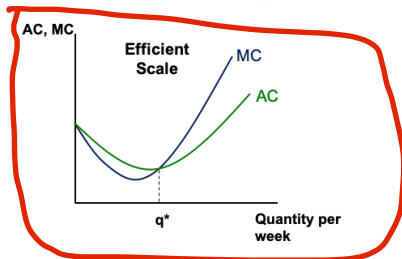
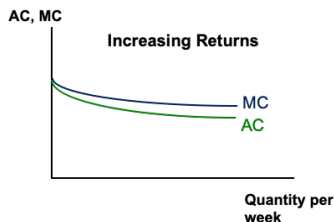
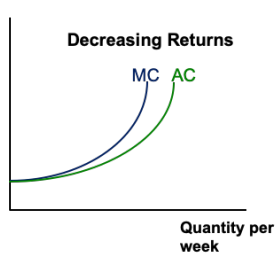
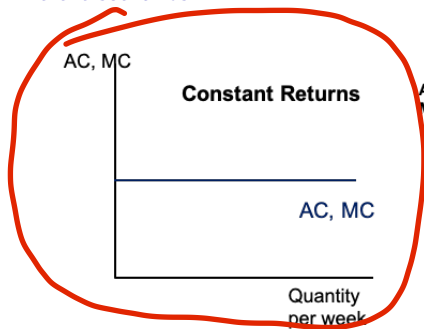
Average and Marginal costs

Different scenarios



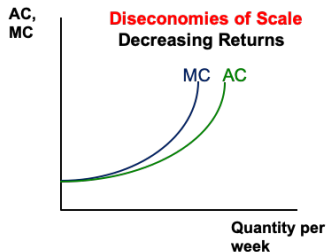
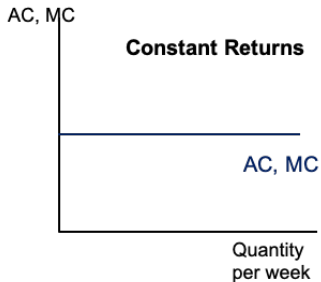
Average and Marginal costs

Different scenarios

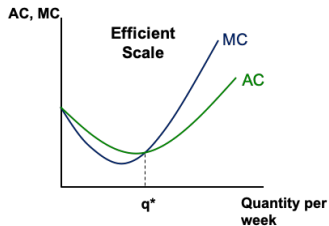
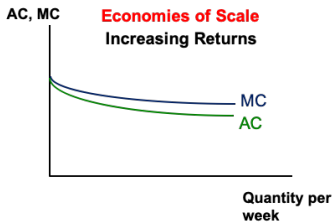
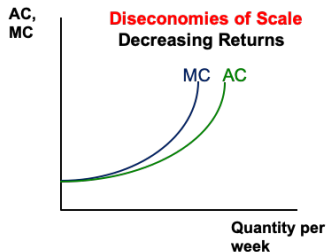
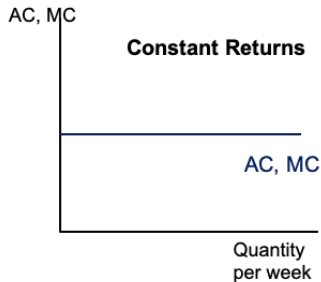


Economies of Scale

Economies of Scale



Economies of Scale



How are *returns to scale* different from *economies of scale*?

Try Yourself!

How are *returns to scale* different from *economies of scale*?

Try Yourself!

Returns to scale refers to how outputs change when all inputs are changed in the same proportion. That is, if output more than doubles when all inputs are doubled, then there are increasing returns to scale.

Economies of scale refers to a property of the cost function – that is whether average costs (AC) rise or fall when more output is produced.

So are these two concepts related?

Relation between *returns to scale* and *economies of scale*

If the production function exhibits increasing returns to scale at all levels of output, then the AC is declining at all points and hence it is a situation with economies of scale.

If the production function exhibits decreasing returns to scale at all levels of output, then the AC is increasing at all points and hence it is a situation with diseconomies of scale.

If the production function exhibits constant returns to scale, then the AC is constant at all points and hence it is neither economies nor diseconomies of scale.

Time-horizon of decision making

Short vs. long run

one of the inputs is fixed.

This denotes the time horizon over which a firm is making decisions.

machines
capital (K) ←
constant
→ (fixed)

↓
mable to change.

Labor = not
fixed
↳ (variable)

L, K
fixed

Time-horizon of decision making

Short vs. long run

This denotes the time horizon over which a firm is making decisions.

For example - if we are to study short run responses to changes in external conditions, then certain inputs (capital) might have to be held constant. If the time interval under consideration is one week, then the size of Tesla's battery manufacturing plant has to be considered constant. However, in the long run, the size of the plant can also change.

In line with this example, in this course we consider the difference between short and long run to be that capital is held constant.

Does this mean firms have no flexibility in short run?

Does this mean firms have no flexibility in short run?

Constrained

No. Although firms can't change the amount of capital, they can change other inputs, which in our case is labor (L). Continuing in the same example, Tesla can hire/fire workers from their factories at relatively short notice (depending on contracts), although the number of car-assembly units is unlikely to change in the short run.

Note that there might be other inputs in the production process that may (or may not) vary in the short run. In this course, we consider the simple two-input model with labor and capital, where capital does not change in the short run.

Types of short-run costs

Types of short-run costs

Fixed costs

These are the costs associated with the fixed factors of production, namely capital (K). These costs are independent of the amount of production – that is, irrespective of how much the firm produces, the fixed costs are ***constant*** in the short run.

Types of short-run costs



long-run

Fixed costs

These are the costs associated with the fixed factors of production, namely capital (K). These costs are independent of the amount of production – that is, irrespective of how much the firm produces, the fixed costs are **constant** in the short run.

Variable costs

These are short-run costs that are associated with the inputs that can be changed in the short run, which in our case is labor (L). In our previous example, although Tesla cannot add new machines to its factory, it could potentially add to its output by hiring more workers – the cost entailed to hire these workers would be a part of the variable costs of production.

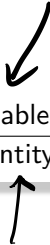
Average costs

New (but related) concepts – short run costs (variable and fixed costs)

$$AC = \frac{TC}{Q}$$

Average costs

New (but related) concepts – short run costs (variable and fixed costs)


$$\text{Average variable cost (AVC)} = \frac{\text{Total variable costs}}{\text{Quantity}} = \frac{TVC}{q}$$

Average costs

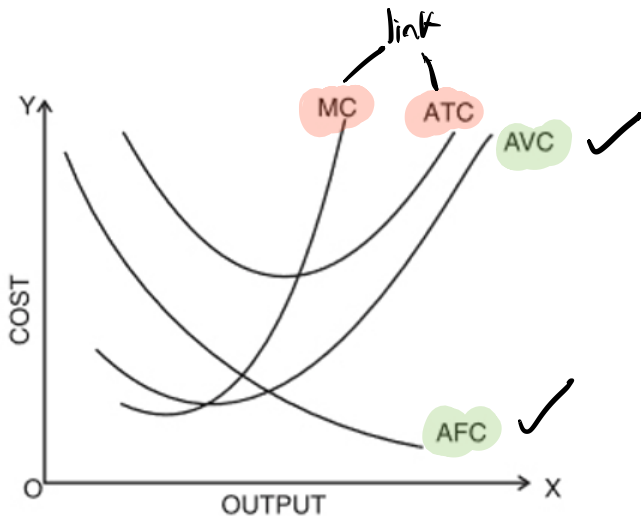
New (but related) concepts – short run costs (variable and fixed costs)

$$\text{Average variable cost (AVC)} = \frac{\text{Total variable costs}}{\text{Quantity}} = \frac{TVC}{q}$$

$$\text{Average fixed cost (AFC)} = \frac{\text{Total fixed costs}}{\text{Quantity}} = \frac{TFC}{q}$$

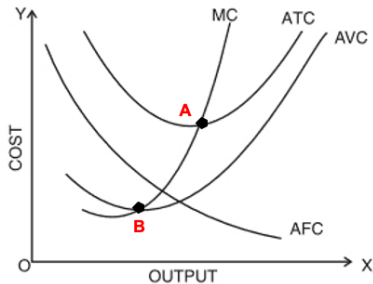
Average fixed and variable costs

Graphical representation



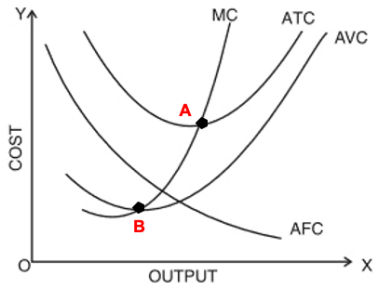
Average fixed and variable costs

Some stylized facts (or properties)



Average fixed and variable costs

Some stylized facts (or properties)



► $ATC = AFC + AVC$ (always true!)

$$TC = TFC + TVC$$

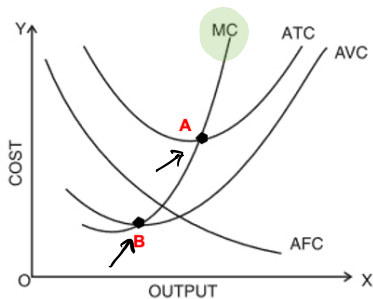
Divide by Q

$$\frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q}$$

$$ATC = AFC + AVC$$

Average fixed and variable costs

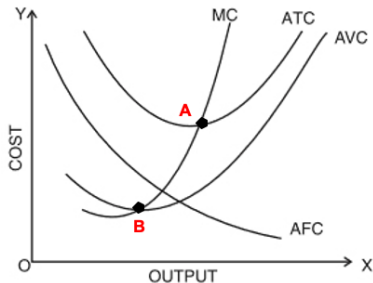
Some stylized facts (or properties)



- ▶ $ATC = AFC + AVC$ (always true!)
- ▶ MC intersects ATC and AVC at their minimum values (points A and B)

Average fixed and variable costs

Some stylized facts (or properties)

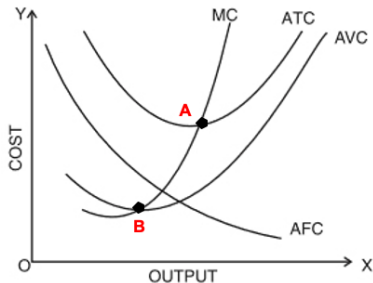


- ▶ $ATC = AFC + AVC$ (always true!)
- ▶ MC intersects ATC and AVC at their minimum values (points A and B)
- ▶ When $MC < AVC$, then AVC is falling (left of point B).

When $MC < AC$, then AC is falling.

Average fixed and variable costs

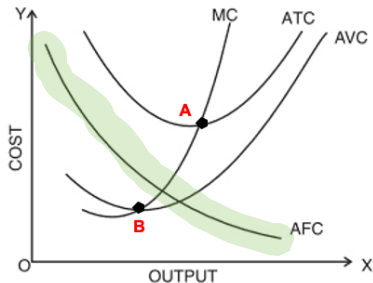
Some stylized facts (or properties)



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- ▶ MC intersects ATC and AVC at their minimum values (points A and B)
- ▶ When $MC < AVC$, then AVC is falling (left of point B).
- ▶ When $MC > AVC$, then AVC is rising (right of point B).

Average fixed and variable costs

Some stylized facts (or properties)

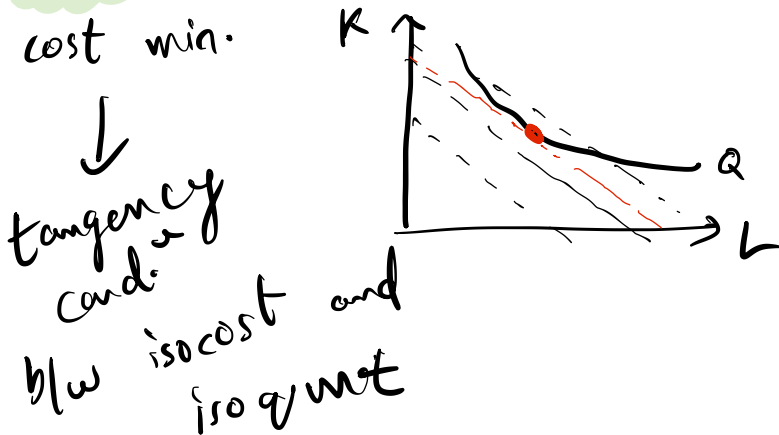


$$AFC = \frac{TFC}{Q} \leftrightarrow Q \uparrow$$

- ▶ $ATC = AFC + AVC$ (always true!)
- ▶ MC intersects ATC and AVC at their minimum values (points A and B)
- ▶ When $MC < AVC$, then AVC is falling (left of point B).
- ▶ When $MC > AVC$, then AVC is rising (right of point B).
- ▶ AFC is always falling. $AFC = \frac{TFC}{q}$.
Since TFC is constant, when q rises then $AFC \downarrow$.

Question: Will a firm always be able to use the cost minimizing combination of L and K in the short run?

Try yourself, and then check out the solution!



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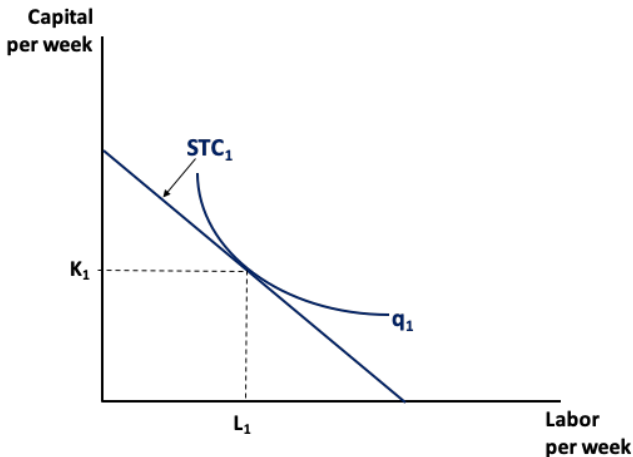
Try yourself, and then check out the solution!

The answer is **No**.

The reason is that the firm does not have the flexibility to choose any level of capital (fixed factor of production) in the short run. It is constrained to stick with the same amount of capital.

Firm decision in the short run

Suppose the firm is currently producing q_1 units of output using L_1 units of labor and K_1 units of capital.

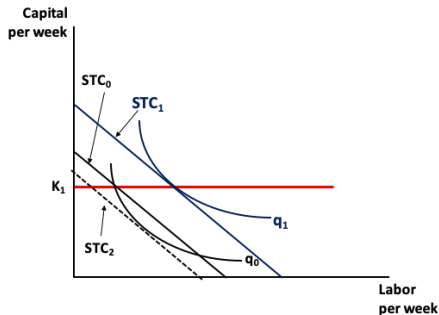


Firm decision in the short run

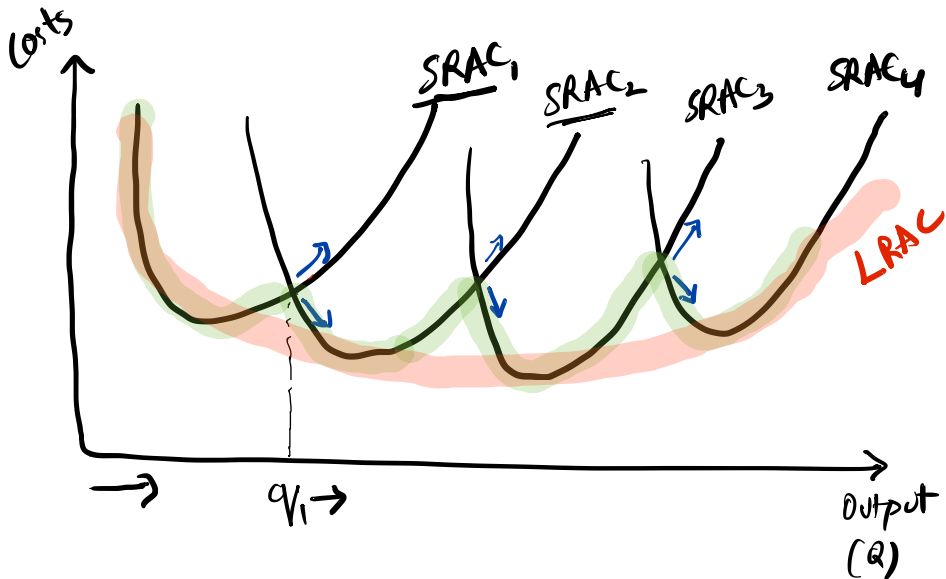
Not always able to choose the cost-minimizing level of inputs.

Now, if the firm wants to produce q_0 units of output, where $q_0 < q_1$. Then the firm will still be using K_1 units of capital (as it is fixed), and will change the units of labor (variable factor) to L_0 .

Note that (L_0, K_1) is not a cost-minimizing input combination for producing output q_0 (tangency condition is not satisfied).

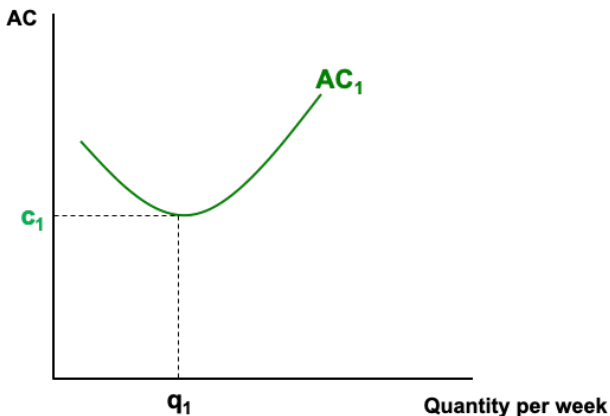


Deriving long run AC from short run AC

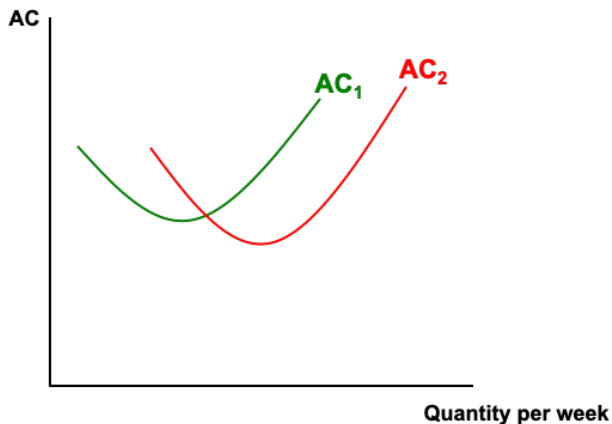


Deriving long run AC from short run AC

In the short run, the level of capital is fixed, so we operate on a given average cost curve. Therefore, the minimum cost of producing quantity q_1 is c_1 .

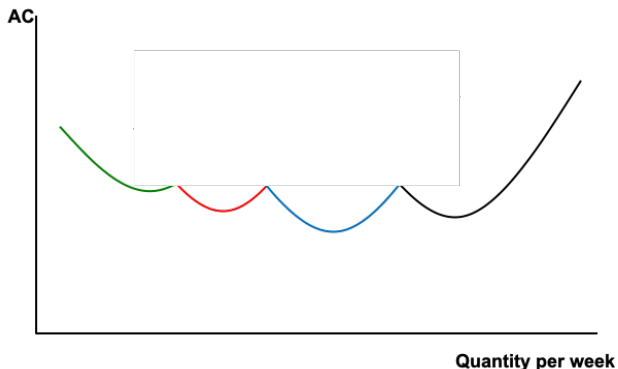


Deriving long run AC from short run AC



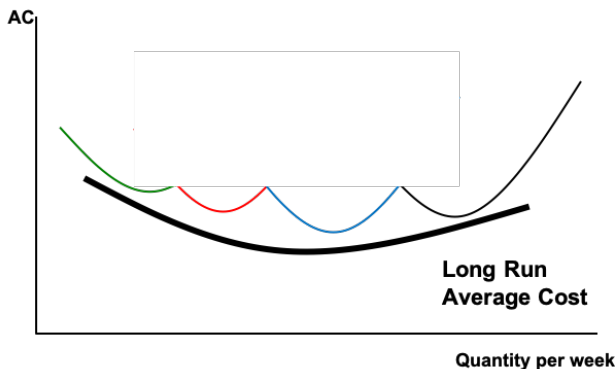
Deriving long run AC from short run AC

Therefore, the long run AC curve is given by the lower envelope of different short run AC curves.



Deriving long run AC from short run AC

Therefore, the long run AC curve is given by the lower envelope of different short run AC curves. In this course, we use a "smoothened" version of this curve.



Deriving long-run total costs $\rightarrow TC, AC, MC.$

Production function $q = K \cdot L$

$w, r = \text{constants}$

$$TC = TC(q)$$

$q = \text{number}$

$$TC = wL + rK$$

optimum L, K for a given q

$$MP_L = \frac{\partial [q]}{\partial L} = \frac{\partial (K \cdot L)}{\partial L} = K$$

$$\frac{MP_L}{MP_K} = \frac{w}{r}$$

$$MP_K = \frac{\partial q}{\partial K} = \frac{\partial (K \cdot L)}{\partial K} = L$$

$$\Rightarrow \frac{K}{L} = \frac{w}{r}$$

$$\Rightarrow L = \frac{K r}{w}$$

$$q = K \cdot L$$

$$\Rightarrow q = K \cdot \left(\frac{K r}{w} \right) \Rightarrow K^2 = \frac{q w}{r}$$

$$\Rightarrow K = \left(\frac{q r w}{r^2} \right)^{1/2}$$

Deriving long-run total costs

Production function: $q = K.L$

$$K = \left(\frac{q}{w}\right)^{1/2} \text{ and } L = \frac{K}{v}$$

$$L = \left(\frac{q}{w}\right)^{1/2} \cdot \frac{1}{v} = \left(\frac{q}{w}\right)^{1/2} \frac{1}{v}$$

$$TC = wL + vk$$

$$= w \left(\frac{q}{w}\right)^{1/2} + v \left(\frac{q}{w}\right)^{1/2}$$

$$= (wq/v)^{1/2} + (wq/v)^{1/2}$$

$$= \underline{\underline{2(wq/v)^{1/2}}} \quad || \quad AC = \frac{TC}{q} = 2\left(\frac{q}{v}\right)^{1/2}$$

Example: Short-run costs

Try it yourself!

A firm producing hockey sticks has the following production function: $q = 2\sqrt{KL}$. In the short run, the firm's capital is fixed at $K = 100$. The rental rate of K is $v = \$1$ and the wage rate for L is $w = \$4$.

- Calculate the firm's short-run total cost and average cost functions.
- The firm's short run marginal cost function is given by $SMC = q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?
- Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).
- Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Example

- a. Calculate the firm's short-run total cost and average cost functions.

Part a.

We know that $q = 2\sqrt{KL}$ and that $K=100$. Therefore,
 $q = 2\sqrt{100L} \implies q = 20\sqrt{L} \implies \sqrt{L} = \frac{q}{20} \implies L = \frac{q^2}{400}$.

Therefore, $STC = vK + wL = 100 + \frac{q^2}{100}$, and

$$SAC = \frac{STC}{q} = \frac{100}{q} + \frac{q}{100}$$

Example

b. The firm's short run marginal cost function is given by $SMC = q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

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We know that $SMC = q/50$

If $q = 25$, then $STC = 106.25$,

Example

b. The firm's short run marginal cost function is given by $SMC = q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

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We know that $SMC = q/50$

If $q = 25$, then $STC = 106.25$, $SAC = 4.25$

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Part b.

We know that $SMC = q/50$

If $q = 25$, then $STC = 106.25$, $SAC = 4.25$ and $SMC = 0.5$

If $q = 50$, then $STC = 125$, $SAC = 2.5$ and $SMC = 1$

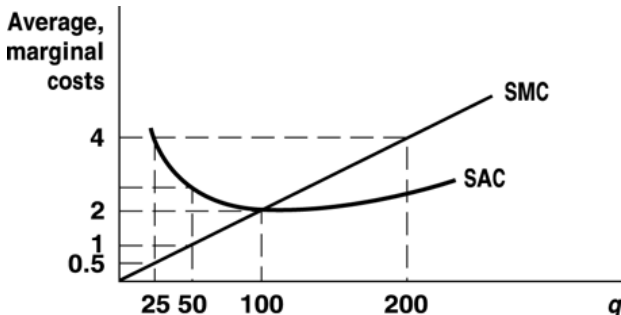
If $q = 100$, then $STC = 200$, $SAC = 2$ and $SMC = 2$

If $q = 200$, then $STC = 500$, $SAC = 2.5$ and $SMC = 4$

Example

c. Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).

Part c.



Example

d. Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Part d.

SMC and SAC intersect at $q = 100$. As long as the marginal cost of producing one more unit is below the average cost curve, average costs will be falling. Similarly, if the marginal cost of producing one more unit is higher than the average cost, then average costs will be rising. Therefore, the SMC curve must intersect the SAC curve at its lowest point.

Deriving long-run total costs

Production function $q = K \cdot L$

$TC, AC, MC = ?$

$w, r = \text{constants.}$

$TC = \min. \text{ cost of producing an output.}$

$TC = TC(q)$ optimum L, K .

$$\frac{MP_L}{w} = \frac{MP_K}{r}$$

$$\frac{K}{w} = \frac{L}{r}$$

$$\frac{K r}{w} = L \quad (1)$$

$$MP_L = \frac{\partial q}{\partial L} = \frac{\partial [K \cdot L]}{\partial L} = K$$

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$$q = K \cdot L = K \cdot \frac{K r}{w} = K^2 \frac{r}{w}$$

$$\Rightarrow K^2 = \frac{q w}{r}$$

$$K = \left[\frac{q w}{r} \right]^{1/2} \quad (2)$$

$$L = K \frac{r}{w} = \left[\frac{q w}{r} \right]^{1/2} \frac{r}{w} \quad (3)$$

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Deriving long-run total costs

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$$TC = wL + rK = w \cdot \left[\frac{q}{r} \right]^{1/2} + r \cdot \left[\frac{q}{w} \right]^{1/2}$$
$$= (q/rw)^{1/2} + (q/rw)^{1/2} = 2[q/rw]^{1/2}$$

$$AC = \frac{TC}{Q} = \frac{2 \left[\frac{Qv + w}{Q} \right]^{1/2}}{Q} = 2 \cdot \left[\frac{v + w}{Q} \right]^{1/2}$$

$$MC = \frac{d\pi}{dq} = \frac{d}{dq} [2(qvw)^{1/2}] = 2 \cdot \frac{1}{2} \left[\frac{vw}{q} \right]^{1/2}$$

$$= \left[\frac{vw}{q} \right]^{1/2} \leftarrow$$

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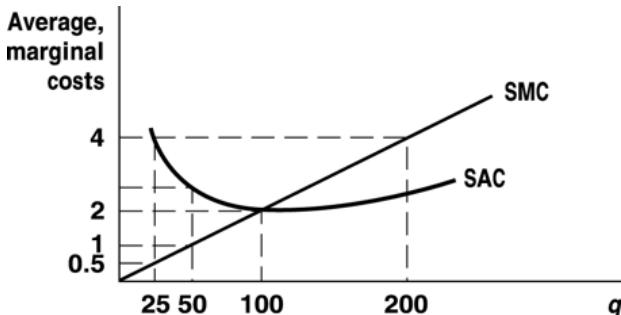
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Example

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